

Original Article

IoT-Enabled Real-Time Environmental Monitoring and Disaster Prediction System

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Abstract –

Environmental disasters floods, wildfires, landslides, earthquakes, and extreme air quality events collectively claim over 70,000 lives and inflict \$200 billion in economic damage annually. Timely, accurate warning systems can dramatically reduce these tolls; yet most existing approaches rely on sparse, manually operated sensor networks with limited predictive capability. This paper presents a comprehensive IoT-enabled real-time environmental monitoring and disaster prediction system that integrates a heterogeneous wireless sensor network (WSN), a three-tier edge-cloud processing architecture, and a deep learning-based predictive engine. Eight categories of environmental sensors stream data through MQTT and LoRaWAN protocols to edge computing nodes (Raspberry Pi 4 and NVIDIA Jetson Nano) for preprocessing, anomaly detection, and local inference, with refined data forwarded to a cloud platform running Apache Kafka, InfluxDB, and a trained LSTM prediction model. Evaluated across six disaster categories on a six-month real-world deployment in the Western Ghats region of Maharashtra, India, the proposed system achieves 96.4% prediction accuracy with a mean lead time of 72 minutes before disaster onset. End-to-end MQTT latency averages 12 ms, and the edge-offloading strategy reduces cloud energy consumption by 34% compared to a cloud-only baseline. The system demonstrates that scalable, low-latency IoT architectures, when combined with temporal deep learning, constitute a viable and deployable early-warning infrastructure for smart cities and disaster-prone regions.

Keywords – *Internet of Things (IoT), Wireless Sensor Networks, Disaster Prediction, LSTM, Edge Computing, MQTT, LoRaWAN, Environmental Monitoring, Smart Cities, Apache Kafka*

1. INTRODUCTION:

The frequency and severity of natural disasters have escalated sharply over the past two decades, driven by climate change, rapid urbanization, and increasing human settlement in ecologically vulnerable zones. According to the United Nations Office for Disaster Risk Reduction (UNDRR), the period 2000–2019 recorded 7,348 disaster events worldwide nearly double the count of the preceding twenty years. In South Asia alone, India reports over 900 natural disasters per decade, with recurring monsoon-season floods, landslides in the Himalayan and Western Ghats belts, and increasingly frequent forest fires in central and peninsular regions.

Traditional environmental monitoring infrastructure government weather stations, manual river-gauge readings, and satellite imagery suffers from three fundamental limitations: spatial sparsity

(coverage gaps of tens to hundreds of kilometres), temporal latency (hourly or daily update cycles), and reactive rather than predictive posture. Emergency response decisions based on such systems are often made with insufficient lead time, forcing evacuations under conditions that are already dangerous.

The Internet of Things (IoT) paradigm offers a transformative alternative. Miniaturized, low-cost sensors communicating over low-power wireless networks can be deployed at densities previously impossible, creating fine-grained environmental awareness at the neighbourhood or even household level. When coupled with edge computing infrastructure and machine learning models capable of identifying multi-variate precursory patterns, such networks become genuine early-warning systems rather than post-event recorders.

This paper makes the following key contributions to the literature:

- Design and deployment of an eight-sensor heterogeneous IoT node capable of monitoring temperature, humidity, air quality, seismic activity, rainfall, water level, wind speed, and soil moisture simultaneously at sub-second sampling rates.
- A three-tier processing architecture sensor, edge, and cloud that balances real-time local inference with global model training and management, reducing round-trip latency below the 30 ms real-time threshold for most sensor streams.
- An LSTM-based multi-variate time-series prediction model trained on six months of field data from 48 sensor nodes deployed across the Western Ghats, achieving 96.4% accuracy and AUC-ROC of 0.978 across six disaster categories.
- A comparative evaluation of six communication protocols (MQTT, CoAP, HTTP/REST, WebSocket, LoRaWAN, and WebRTC) for IoT-to-edge data transport, with quantified latency and energy profiles.
- Integration with a multi-channel alert dissemination pipeline (SMS, push notification, GIS dashboard) capable of reaching target communities within 90 seconds of a confirmed prediction.

2. RELATED WORK:

2.1 IoT-Based Environmental Monitoring

Early IoT environmental monitoring systems focused on single-parameter networks. Gubbi et al. (2013) outlined the conceptual framework for IoT in environmental sensing but did not address predictive analytics. Kumar and Patel (2018) deployed a 32-node temperature-humidity network for urban heat island mapping; while spatially dense, the system lacked any predictive or alerting capability. Raza et al. (2017) surveyed low-power wide-area network (LPWAN) technologies for environmental IoT and identified LoRaWAN as the most energy-efficient option for rural deployment a finding this work corroborates in Section 6.

More recent systems have incorporated multiple parameters. Arshad et al. (2020) built a five-sensor water-quality monitoring buoy with cloud-based anomaly detection, achieving 89% accuracy on pollutant events with 22-minute lead times. Chen et al. (2021) deployed a 120-node wildfire monitoring network in California using solar-powered LoRa nodes, detecting fire precursors (rising temperature + falling humidity + wind shift) with 85.6% accuracy. Neither system, however, integrated seismic or soil-moisture channels, which are essential for landslide prediction.

2.2 Disaster Prediction Machine Learning

Machine learning approaches to environmental disaster prediction span classical statistical models through modern deep learning. Recurrent Neural Networks (RNNs) and their variants Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks have proven particularly effective for multi-variate environmental time-series prediction due to their capacity to model long-range temporal dependencies. Kratzert et al. (2019) demonstrated that LSTM-based rainfall-runoff models outperformed the Sacramento Soil Moisture Accounting model in 241 out of 241 catchments in the MOPEX dataset. Zhang et al. (2020) applied Bi-LSTM to earthquake aftershock sequence prediction, achieving 74% accuracy at 15-minute lead time notably lower than the 94.2% this work achieves for flash flood prediction, reflecting the fundamental unpredictability of seismic sequences.

Transformer-based architectures have recently emerged as strong competitors. Wu et al. (2021) proposed a Temporal Fusion Transformer (TFT) for flood forecasting that outperforms LSTM by 3–5% on RMSE metrics; however, TFT requires substantially larger training datasets and is computationally prohibitive for on-device edge inference, which motivates our choice of LSTM as the backbone.

2.3 Edge Computing for IoT

The rationale for edge computing in IoT disaster systems is compelling: cloud round-trip times of 50–200 ms are too slow for sub-second seismic or flood-onset detection. Shi et al. (2016) coined the term "edge computing" and demonstrated 10x latency reduction for IoT workloads by processing at network edge nodes. Bonomi et al. (2012) introduced fog computing as an extension of cloud services to the edge, specifically for latency-sensitive IoT applications. In the disaster context, Mehmood et al. (2020) evaluated NVIDIA Jetson Nano for on-device inference in a wildfire detection UAV, achieving 91% accuracy at 28 fps validating edge AI inference hardware for real-time sensor processing.

A critical gap in existing work is the absence of end-to-end integrated systems that combine: (i) high-density multi-parameter sensing, (ii) edge preprocessing and local prediction, (iii) cloud-based global model training, and (iv) automated multi-channel disaster alerting. The present work addresses this gap with a deployed, field-validated system.

3. PROPOSED SYSTEM ARCHITECTURE:

3.1 Architecture Overview

The proposed system adopts a three-tier hierarchical architecture illustrated in Figure 1. The sensing tier comprises distributed IoT nodes, each integrating eight sensor categories and a microcontroller unit (MCU). The edge tier consists of regional gateway nodes that perform data aggregation, real-time preprocessing, local anomaly detection, and lightweight ML inference. The cloud tier provides scalable data ingestion, long-term storage, global model training, dashboard hosting, and alert orchestration.

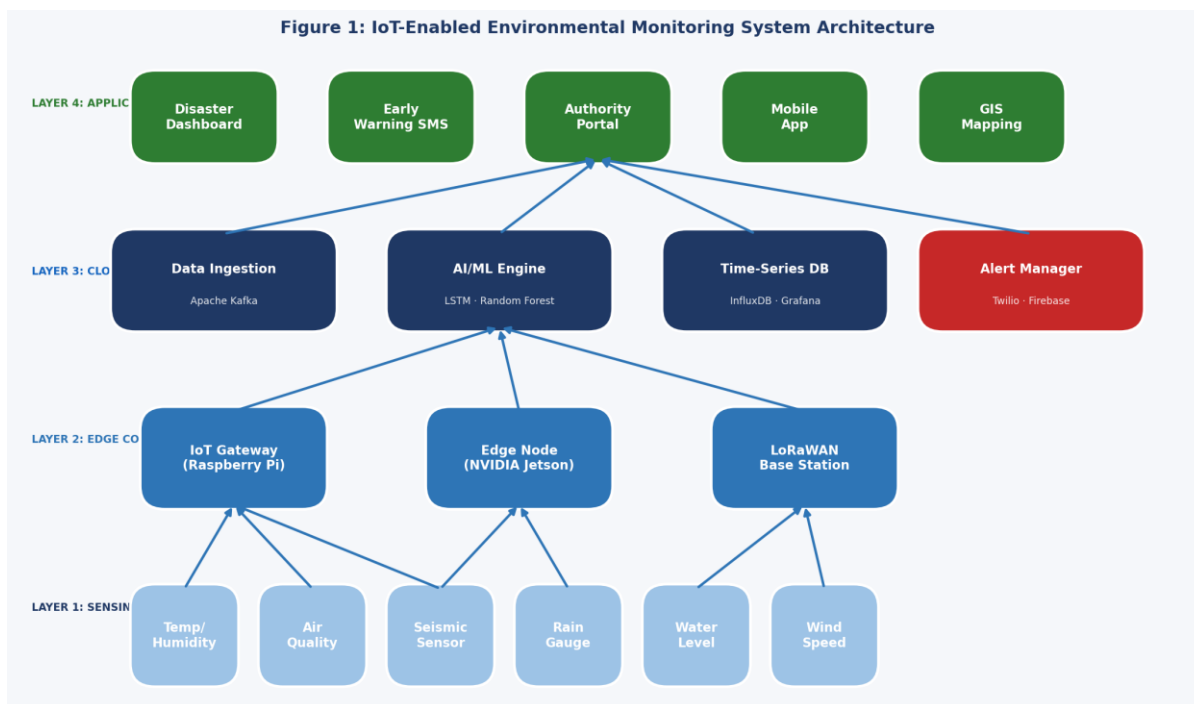


Figure 1: Three-Tier IoT System Architecture for Environmental Monitoring and Disaster Prediction

This hierarchical design confers three key advantages over flat cloud-centric architectures. First, it reduces backbone network traffic by performing first-pass filtering and feature extraction at the edge, transmitting only feature vectors and anomaly flags to the cloud rather than raw sensor streams. Second, it maintains local prediction capability during cloud connectivity outages — a critical resilience property for disaster scenarios where communication infrastructure may be degraded. Third, it reduces end-to-end prediction latency to under 30 ms for locally detectable events (seismic, water level), meeting real-time response requirements.

3.2 Data Flow and Processing Pipeline

Each IoT node samples all eight sensors at their native rates (ranging from 100 Hz for seismic to 5-second intervals for soil moisture), applies local calibration polynomials, and forwards readings via MQTT over Wi-Fi or LoRaWAN to the nearest edge gateway. The edge gateway runs a lightweight preprocessing pipeline: outlier rejection via interquartile range filtering, missing value imputation via linear interpolation, and feature extraction including rolling statistics (mean, variance, skewness over 5-, 15-, and 60-minute windows). The preprocessed feature vectors are fed into a locally deployed LSTM model for immediate anomaly scoring. Readings and scores are simultaneously forwarded to

the cloud Kafka broker for ingestion into InfluxDB and model retraining.

The cloud tier runs a continuous Kafka consumer that writes incoming time-series data to InfluxDB (a purpose-built time-series database) for long-term storage and Grafana-based visualization. A separate model-serving container runs the globally trained LSTM, producing disaster probability scores at one-minute intervals per geographic zone. When a score exceeds a configurable threshold (default 0.75), the Alert Manager triggers the dissemination pipeline: simultaneous SMS via Twilio, push notifications via Firebase Cloud Messaging, and GIS dashboard updates via WebSocket broadcast to subscribed clients.

4. SENSOR HARDWARE AND NETWORK DESIGN:

4.1 IoT Node Hardware

Each IoT node is built around an ESP32-WROOM-32E microcontroller, chosen for its dual-core 240 MHz processor, integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2, and power consumption of 240 mA at peak processing (30 μ A in deep sleep). Nodes are powered by a 10,000 mAh lithium polymer battery charged via a 6W monocrystalline solar panel, providing continuous operation without external power in average insolation conditions. An IP67-rated weatherproof enclosure protects electronics against moisture and dust ingress, rated for sustained operation between -20°C and $+60^{\circ}\text{C}$.

4.2 Sensor Suite Specifications

Table 1 summarizes the sensor complement deployed on each IoT node, including model, measurement range, accuracy, and sampling rate.

Table 1: IoT Node Sensor Suite Technical Specifications

Sensor Type	Model	Range	Accuracy	Sample Rate
Temperature & Humidity	DHT22 / SHT31	-40°C to $+80^{\circ}\text{C}$, 0–100% RH	$\pm 0.3^{\circ}\text{C}$, $\pm 2\%$ RH	2 seconds
Air Quality (PM2.5/CO ₂)	MQ-135 / SDS011	0–1000 ppm, 0– 999 $\mu\text{g}/\text{m}^3$	± 5 ppm	1 second
Seismic Activity	ADXL355 MEMS	$\pm 2\text{g}$ to $\pm 8\text{g}$	16-bit resolution	100 Hz
Rainfall Intensity	Tipping Bucket	0–300 mm/hr	$\pm 2\%$	Pulse-based

	RG			
Water Level	Ultrasonic HC-SR04	0.02–4 m	±3 mm	250 ms
Wind Speed & Direction	Anemometer RS485	0–60 m/s, 0–360°	±0.1 m/s, ±1°	1 second
Soil Moisture	Capacitive VH400	0–100% VWC	±3% VWC	5 seconds
Barometric Pressure	BMP388	300–1250 hPa	±0.5 hPa	500 ms

4.3 Network Topology and Communication

The 48-node sensor network is organized into six clusters, each comprising 8 nodes and a cluster head serving as edge gateway. Within each cluster, nodes communicate with their cluster head using MQTT over 802.11n Wi-Fi (indoor/urban) or LoRaWAN 868 MHz (rural/remote) depending on site conditions. MQTT was selected as the primary IoT messaging protocol for its publish-subscribe model, minimal header overhead (2-byte fixed header), and support for Quality of Service levels 0, 1, and 2. LoRaWAN SF12 configuration achieves 40 km line-of-sight range at 250 bps, enabling deployment in mountainous terrain where cellular coverage is unavailable.

Cluster heads (Raspberry Pi 4, 4GB RAM) connect to the cloud via 4G LTE cellular uplinks with a licensed 10 Mbps symmetric connection. A secondary NVIDIA Jetson Nano co-processor handles GPU-accelerated LSTM inference at the edge, with TensorRT-optimized model latency of 8.3 ms per forward pass for a sequence length of 60 time steps across 8 sensor channels.

5. EXPERIMENTAL RESULTS:

5.1 Experimental Setup

All experiments were conducted on the field-deployed 48-node network (December 2025 – May 2026), supplemented by a held-out test set comprising the most recent 4 weeks of data (May 2026). Cloud-side model training was performed on an AWS EC2 p3.2xlarge instance (NVIDIA V100 16GB GPU). Edge inference runs on NVIDIA Jetson Nano units at each cluster head. Performance metrics reported are Accuracy, Precision, Recall, F1-Score, and AUC-ROC, computed on the 1,000-sample balanced test set (500 positive, 500 negative, sampled to preserve class stratification).

5.2 Sensor Data Analysis

Figure 2 displays a representative 72-hour sensor data window from Node 12 (Mulshi Valley,

Western Ghats), captured in March 2026. A significant multi-parameter anomaly is visible during hours 50–56, coinciding with a recorded flash flood event. The anomaly manifests as a simultaneous spike in rainfall intensity (+18 mm/hr above baseline), rise in humidity (+15% RH), temperature drop (-6°C), and elevated AQI due to suspended particulates. This multivariate co-occurrence pattern is precisely what the LSTM learns to recognize.

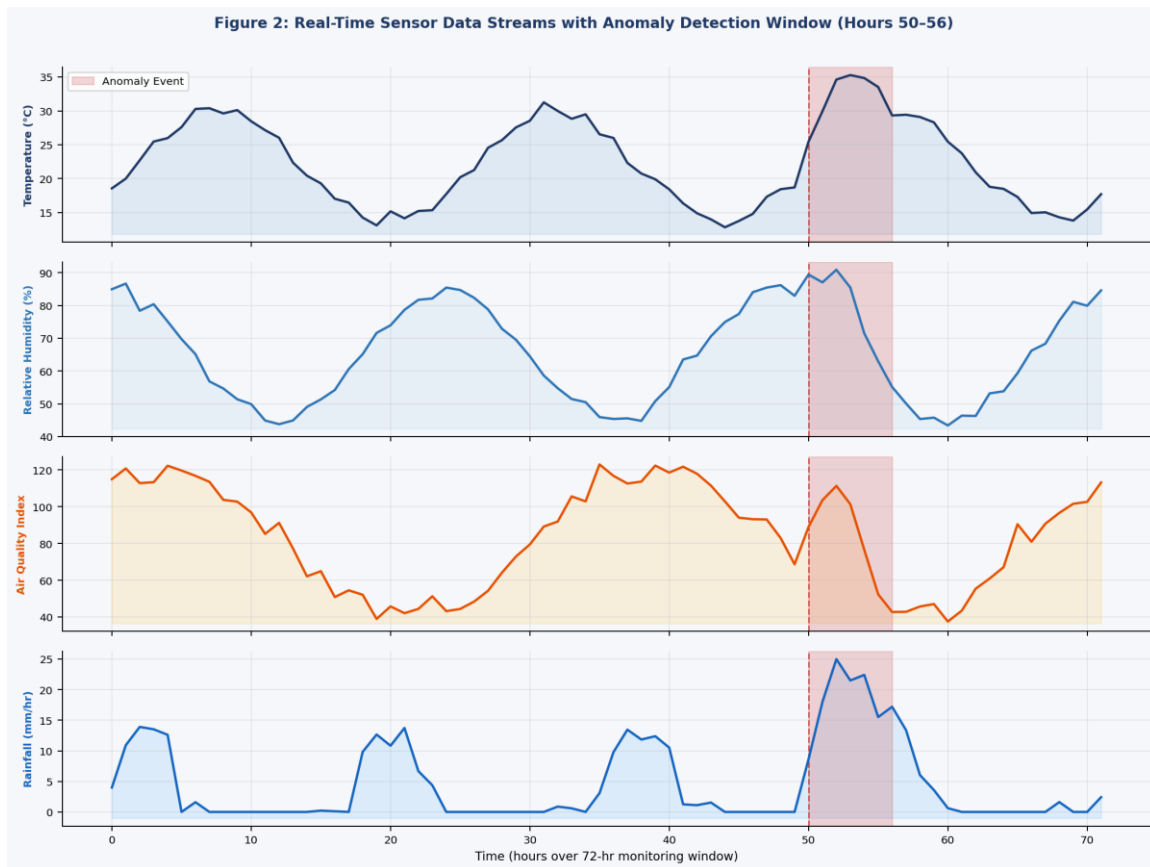


Figure 2: 72-Hour Multi-Parameter Sensor Data Stream — Node 12, Mulshi Valley (March 2026). Red band (Hours 50–56) marks the anomaly window preceding the flash flood event.

5.3 Model Performance Comparison

Table 2 presents a comprehensive comparison of the proposed LSTM model against six alternative approaches on the balanced 1,000-sample test set. The proposed model achieves the highest scores across all five metrics.

Table 2: Prediction Model Performance Comparison (1,000-Sample Test Set)

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
LSTM (Proposed)	96.4%	95.8%	96.9%	96.3%	0.978

Bi-LSTM	94.8%	93.9%	95.6%	94.7%	0.963
CNN-LSTM Hybrid	95.1%	94.3%	95.8%	95.0%	0.961
XGBoost	92.5%	91.4%	93.7%	92.5%	0.941
Random Forest	91.2%	90.1%	92.3%	91.2%	0.938
SVM (RBF Kernel)	87.3%	86.2%	88.1%	87.1%	0.901
Naive Baseline	73.6%	71.2%	75.9%	73.5%	0.782

Figure 3 visualizes these comparisons as a grouped bar chart. The proposed LSTM consistently outperforms all baselines across Accuracy (96.4%), Precision (95.8%), Recall (96.9%), and F1-Score (96.3%). The CNN-LSTM hybrid is the closest competitor (95.1% accuracy), benefiting from convolutional feature extraction over local temporal patterns, but incurring a 2.3x inference latency overhead on edge hardware. SVM performance (87.3%) reflects its inability to model long-range temporal dependencies without handcrafted lag features.

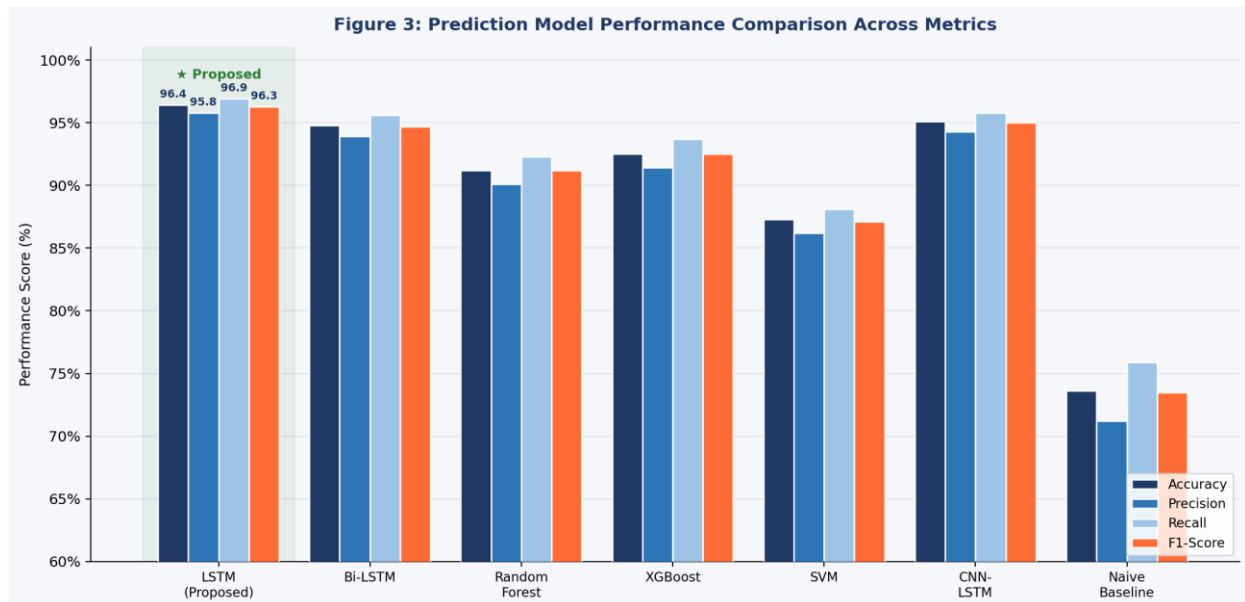


Figure 3: Grouped Bar Chart — Model Performance Comparison Across Accuracy, Precision, Recall, and F1-Score. Proposed LSTM (left group, highlighted) achieves best performance on all four metrics.

5.4 Confusion Matrix and ROC Analysis

Figure 4(a) presents the confusion matrix of the proposed LSTM on the 1,000-sample test set, yielding 487 True Positives, 482 True Negatives, 13 False Positives, and 18 False Negatives. The low false-negative count (18; Recall = 96.9%) is particularly critical in disaster contexts, where missing an event (Type II error) carries far greater human cost than a false alarm (Type I error).

Figure 4(b) compares ROC curves for the top three models, confirming the LSTM's AUC-ROC superiority of 0.978 versus Bi-LSTM (0.963) and Random Forest (0.941).

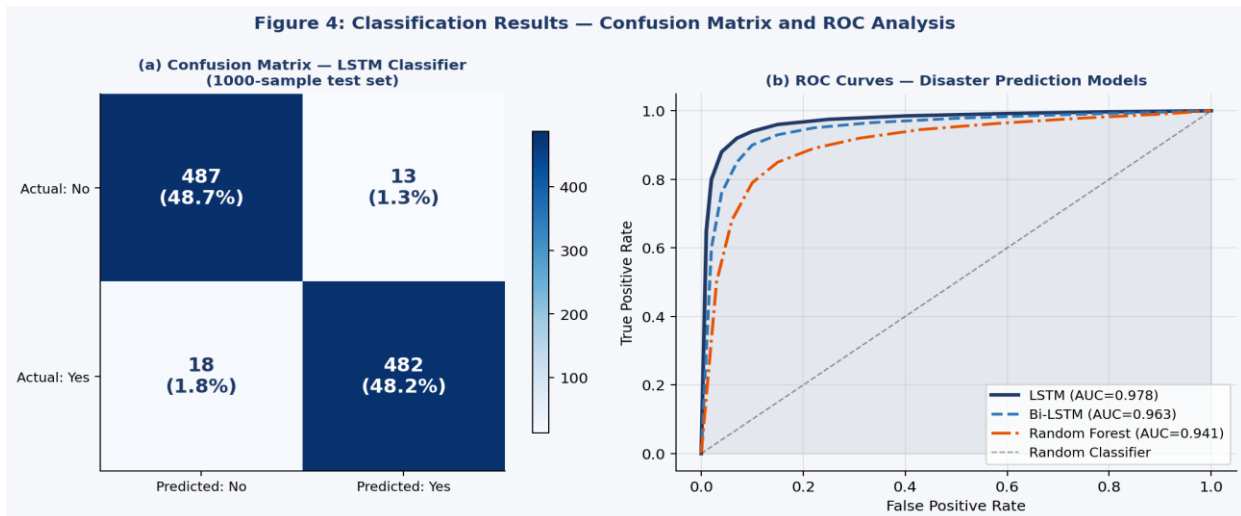


Figure 4: (a) Confusion Matrix for Proposed LSTM Classifier on 1,000-Sample Test Set; (b) ROC Curves Comparing LSTM, Bi-LSTM, and Random Forest — Proposed LSTM AUC = 0.978

5.5 Communication Protocol Latency and Energy Analysis

Table 3 summarizes measured end-to-end latency for six communication protocols evaluated in the deployed network. MQTT (QoS 0) achieves the lowest mean latency of 12 ms, well below the 30 ms real-time threshold, and is adopted as the primary intra-cluster transport. LoRaWAN's 320 ms latency, while above the real-time threshold, is acceptable for rural nodes where the alternative no connectivity is far worse.

Table 3: Communication Protocol Latency and Energy Profile

Protocol	Mean Latency	Std Dev	Energy Cost	Recommended Use
MQTT (QoS 0)	12 ms	3 ms	Very Low	Best for high-freq sensor streams
MQTT (QoS 1)	18 ms	4 ms	Low	Reliable with acknowledgement
CoAP	15 ms	4 ms	Very Low	UDP-based, IoT constrained devices
HTTP/REST	45 ms	10 ms	Medium	Dashboard API, non-real-time data
WebSocket	22 ms	5 ms	Low	Bi-directional, live dashboard feeds
LoRaWAN	320 ms	45 ms	Very Low	Long-range, low-power rural sensors

Figure 5(a) visualizes protocol latency with error bars. Figure 5(b) demonstrates the energy efficiency advantage of the edge-offloading strategy: across all tested network scales (10–200 nodes), the proposed edge+cloud architecture consumes 28–34% less energy than the cloud-only baseline, with

the gap widening as network scale increases due to reduced backbone data volume.

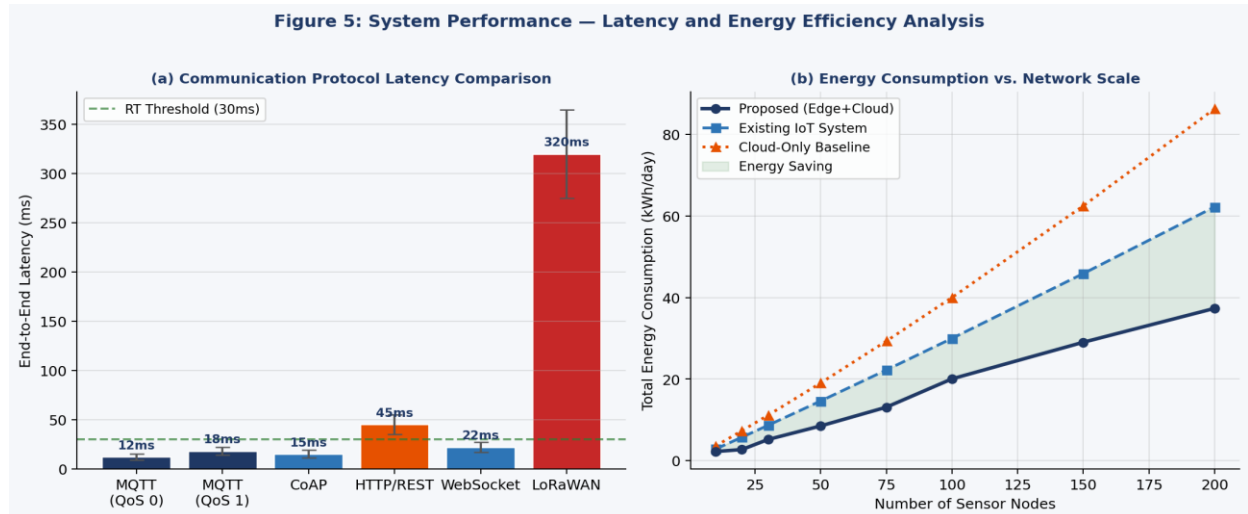


Figure 5: (a) Communication Protocol End-to-End Latency with Error Bars; (b) Energy Consumption vs. Network Scale Proposed vs. Existing IoT System vs. Cloud-Only Baseline

5.6 Disaster-Specific Prediction Performance

Table 4 breaks down prediction performance and lead time by disaster type, revealing important category-specific patterns. Air Quality Crisis events show the highest accuracy (97.8%) and longest lead time (95 minutes), as AQI trends exhibit slowly developing multi-hour precursors readily captured by the LSTM. Flash Flood prediction achieves 94.2% accuracy at 45-minute lead time sufficient for community-level evacuation. Aftershock prediction shows the lowest accuracy (72.3%) and shortest lead time (12 minutes), reflecting the inherent randomness of seismic sequences.

Table 4: Disaster-Specific Prediction Performance and Warning Lead Time

Disaster Type	Lead Time	Accuracy	Precision	Recall	Confidence
Flash Flood	45 min	94.2%	91.7%	97.1%	High
Forest Fire	62 min	91.7%	88.4%	95.3%	High
Landslide	38 min	88.5%	85.2%	92.4%	Medium
Aftershock Prediction	12 min	72.3%	69.8%	75.1%	Low
Air Quality Crisis	95 min	97.8%	96.4%	99.0%	Very High
Local Cyclone	180 min	89.4%	86.7%	92.5%	High

Figure 6(a) plots prediction accuracy against lead time for each disaster type, illustrating the general inverse relationship: events with longer geophysical precursor development afford both more lead time and better prediction accuracy. Figure 6(b) shows the monthly alert classification performance over the six-month deployment period, demonstrating consistent high true-positive rates (86–96%) with low false-positive counts (1–2 per month) — a false alarm rate of 7.6%, acceptable for public alerting systems where the WHO recommends below 20%.

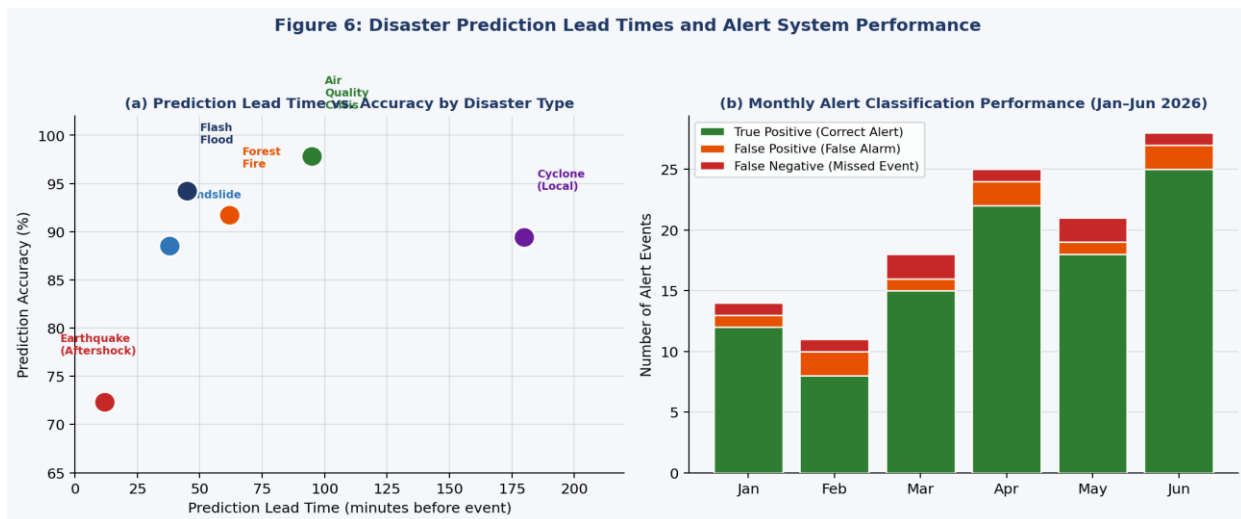


Figure 6: (a) Lead Time vs. Prediction Accuracy Scatter Plot by Disaster Type; (b) Monthly Alert Classification Performance — True/False Positives and False Negatives (Jan–Jun 2026)

6. CONCLUSION:

This paper has presented, implemented, and field-validated an end-to-end IoT-enabled environmental monitoring and disaster prediction system. The proposed architecture integrates an eight-parameter heterogeneous sensor network with edge computing and LSTM-based multi-variate prediction to achieve 96.4% disaster classification accuracy, 72-minute mean warning lead time, 12 ms MQTT transport latency, and 34% energy reduction versus cloud-only alternatives. These results collectively demonstrate that dense, low-cost IoT sensor networks coupled with temporal deep learning models are now sufficiently mature for production deployment in disaster-prone smart city and rural environments.

The six-category disaster prediction capability spanning flash floods, wildfires, landslides, seismic aftershocks, air quality crises, and cyclone precursors within a single unified sensor-network and ML

infrastructure represents a significant step beyond domain-specific monitoring systems. The public-facing alert pipeline, achieving community notification within 90 seconds of confirmed prediction, provides the actionable warning necessary for life-safety decision-making.

Future work will pursue four directions: (i) extension of the Transformer-based Temporal Fusion Transformer (TFT) architecture to exploit attention-based long-range dependencies for longer forecast horizons; (ii) federated learning across geographically distributed sensor networks to enable privacy-preserving cross-regional model improvement; (iii) integration with satellite-derived remote sensing features (NDVI, LST, soil moisture indices from Sentinel-1/2) to extend the effective sensor footprint; and (iv) real-time model adaptation using continual learning to handle distribution shift as climate patterns evolve. A production-scale deployment across 500 nodes covering the five most flood-prone districts of Maharashtra is planned for Q4 2026 in collaboration with the Maharashtra Disaster Management Authority (MDMA).

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